LAPORAN PRAKTIKUM IMPLEMENTASI VISI KOMPUTER TERAPAN: ANALISIS OBJEK, PENGINDERAAN JAUH, CITRA MEDIS, DAN SISTEM PENGAWASAN

PENGOLAHAN CITRA DIGITAL

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A. Pendahuluan

Visi komputer (*computer vision*) merupakan cabang kecerdasan buatan yang memungkinkan mesin untuk mengekstraksi informasi bermakna dari citra digital maupun video. Laporan ini merangkum empat penerapan utama visi komputer dalam skenario dunia nyata: deteksi objek umum, penginderaan jauh untuk ekologi, diagnostik anomali medis, serta sistem pengawasan keamanan cerdas. Pendekatan yang digunakan mencakup pemrosesan citra digital klasik hingga pemanfaatan model pra-latih (*pre-trained model*) berbasis *Deep Learning*.

B. Metodologi dan Implementasi

1. Deteksi Objek Menggunakan YOLO (Praktikum 7.1)

Modul ini berfokus pada pengenalan dan lokalisasi objek menggunakan algoritma YOLO (*You Only Look Once*). YOLO dipilih karena kemampuannya memproses citra secara *real-time* dengan akurasi yang tinggi.

- **Mekanisme:** Implementasi dilakukan memanfaatkan modul *Deep Neural Network* (DNN) dari OpenCV. Citra masukan dikonversi menjadi *blob* agar sesuai dengan arsitektur jaringan saraf YOLOv3.
- **Pemrosesan Hasil:** Output dari jaringan berupa kotak pembatas (*bounding boxes*), metrik kepercayaan (*confidence score*), dan identitas kelas. Untuk mencegah tumpang tindih deteksi pada satu objek yang sama, teknik *Non-Maximum Suppression* (NMS) diterapkan untuk memfilter kotak dengan probabilitas lebih rendah.

2. Analisis Vegetasi Penginderaan Jauh dengan NDVI (Praktikum 7.2)

Pada bidang penginderaan jauh (*remote sensing*), visi komputer digunakan untuk menganalisis tutupan lahan berdasarkan pantulan spektral. Fokus utama adalah ekstraksi *Normalized Difference Vegetation Index* (NDVI).

- **Pendekatan Spektral:** Program menyimulasikan data citra satelit multispektral yang terdiri dari pita spektrum Merah (*Visible Light*) dan Inframerah Dekat (*Near-Infrared*/NIR).
- **Kalkulasi Indeks:** NDVI dihitung menggunakan persamaan:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

- **Klasifikasi:** Nilai NDVI berada pada rentang -1 hingga 1. Rentang ini kemudian disegmentasi untuk mengklasifikasikan jenis tutupan lahan, di mana nilai negatif merepresentasikan badan air, nilai rendah menunjukkan

tanah kosong/perkotaan, dan nilai positif tinggi mengindikasikan vegetasi yang lebat dan sehat.

3. **Deteksi Anomali pada Citra Medis (Praktikum 7.3)**

   Analisis citra medis mensyaratkan presisi tingkat tinggi untuk mendeteksi abnormalitas secara otomatis. Tiga modalitas citra disimulasikan dan diproses menggunakan metode analisis yang disesuaikan dengan karakteristik masing-masing anomali:

   - **Citra X-Ray (Deteksi Fraktur):** Retakan pada tulang memiliki karakteristik garis yang tegas. Oleh karena itu, deteksi dilakukan dengan peningkatkan kontras (Ekualisasi Histogram), ekstraksi fitur tepi (*Canny Edge Detection*), dan identifikasi garis lurus menggunakan *Hough Line Transform*.
   - **Citra MRI (Deteksi Tumor):** Jaringan tumor seringkali muncul sebagai area hiperintens (lebih terang). Proses segmentasi menggunakan *Thresholding* biner, yang diikuti oleh operasi Morfologi Terbuka dan Tertutup (*Opening/Closing*) untuk mereduksi *noise*, lalu kontur diekstraksi dan disaring berdasarkan area dan irregularitas bentuknya.
   - **Citra Retina (Diabetic Retinopathy):** Mikroaneurisma berwujud titik-titik gelap berskala sangat kecil. Filter *Difference of Gaussians* (DoG) diterapkan untuk menonjolkan fitur bintik (*spot detection*), sebelum dianalisis lebih lanjut menggunakan segmentasi kontur.

4. **Sistem Pengawasan: Penghitungan dan Pelacakan Objek (Praktikum 7.4)**

   Sistem pengawasan cerdas (*surveillance*) dirancang untuk mendeteksi gerakan dan melacak lintasan orang (*people tracking*) dalam rekaman video.

   - **Deteksi Gerakan:** Dua metode dikomparasikan. Pertama, *Background Subtraction*, yang mencari selisih piksel absolut antara *frame* referensi (latar belakang) dan *frame* saat ini. Kedua, *Optical Flow* (algoritma Farneback), yang menganalisis vektor arah dan magnitudo perpindahan setiap piksel.
   - **Penghitungan dan Pelacakan:** Hasil dari deteksi gerakan diubah menjadi masker biner. *Blob detection* digunakan untuk membedakan dan menghitung jumlah manusia berdasarkan rentang area piksel. Data historis koordinat objek disimpan secara temporal lintas-*frame* untuk menyimulasikan lintasan (*trajectory*) pergerakan objek.

C. Kesimpulan

Dari serangkaian praktikum ini, dapat disimpulkan bahwa tidak ada satu algoritma visi komputer yang "mutlak" untuk segala kondisi. Pemecahan masalah visual sangat bergantung pada konteks spasial dan karakteristik fitur target. *Deep Learning* (seperti YOLO) sangat unggul untuk pengenalan objek semantik, sementara teknik pemrosesan citra spasial klasik (seperti *thresholding*, indeks spektral, dan analisis morfologi) tetap menjadi metode yang sangat efisien, ringan secara komputasi, dan efektif untuk deteksi fitur terukur pada penginderaan jauh dan citra medis. Analisis gerakan adaptif seperti *Optical Flow* terbukti esensial dalam mengekstraksi data dinamis pada sistem pengawasan pintar.

D. Lampiran Kode Program dan Output

1. Kode Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import requests
from io import BytesIO
import torch
import torchvision
from PIL import Image

def praktikum_7_1():
    print("COMPUTER VISION: OBJECT DETECTION WITH YOLO")
    print("=" * 50)

    # Download sample image
    def download_sample_image():
        # URL diperbaiki: Menggunakan gambar sampel standar YOLO
        url = "https://raw.githubusercontent.com/pjreddie/darknet/master/data/dog.jpg"
        response = requests.get(url)
        img = Image.open(BytesIO(response.content))
        return cv2.cvtColor(np.array(img), cv2.COLOR_RGB2BGR)

    # Load YOLO model (using OpenCV's DNN module)
    def load_yolo_model():
        # Download YOLO weights and config
        weights_url = "https://github.com/pjreddie/darknet/raw/master/cfg/yolov3.weights"
        config_url = "https://github.com/pjreddie/darknet/raw/master/cfg/yolov3.cfg"
        names_url = "https://github.com/pjreddie/darknet/raw/master/data/coco.names"

        # Load class names
        response = requests.get(names_url)
        classes = response.text.strip().split('\n')

        # Load model
        net = cv2.dnn.readNetFromDarknet(config_url, weights_url)
        net.setPreferableBackend(cv2.dnn.DNN_BACKEND_OPENCV)
        net.setPreferableTarget(cv2.dnn.DNN_TARGET_CPU)

        return net, classes

    # Object detection function
    def detect_objects_yolo(net, image, classes, conf_threshold=0.5, nms_threshold=0.4):
        height, width = image.shape[:2]

        # Create blob from image
        blob = cv2.dnn.blobFromImage(image, 1/255.0, (416, 416), swapRB=True, crop=False)
        net.setInput(blob)

        # Get output layer names
        layer_names = net.getLayerNames()
        # Perbaikan kompatibilitas OpenCV Colab: tambahkan .flatten()
        output_layers = [layer_names[i - 1] for i in net.getUnconnectedOutLayers().flatten()]

        # Forward pass
        outputs = net.forward(output_layers)

        # Process detections
        boxes = []
        confidences = []
        class_ids = []

        for output in outputs:
            for detection in output:
                scores = detection[5:]
                class_id = np.argmax(scores)
                confidence = scores[class_id]

                if confidence > conf_threshold:
                    center_x = int(detection[0] * width)
                    center_y = int(detection[1] * height)
                    w = int(detection[2] * width)
                    h = int(detection[3] * height)

                    x = int(center_x - w/2)
                    y = int(center_y - h/2)

                    boxes.append([x, y, w, h])
                    confidences.append(float(confidence))
                    class_ids.append(class_id)

        # Apply non-maximum suppression
        indices = cv2.dnn.NMSBoxes(boxes, confidences, conf_threshold, nms_threshold)

        # Draw detections
        result_image = image.copy()
        if len(indices) > 0:
            for i in indices.flatten():
                x, y, w, h = boxes[i]
                label = f"{classes[class_ids[i]]}: {confidences[i]:.2f}"

                # Draw bounding box
                cv2.rectangle(result_image, (x, y), (x + w, y + h), (0, 255, 0), 2)
                # Draw label
                cv2.putText(result_image, label, (x, y - 10),
                            cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 2)

        return result_image, len(indices)

    try:
        # Load model and image
        net, classes = load_yolo_model()
        image = download_sample_image()

        # Perform detection
        result_image, num_detections = detect_objects_yolo(net, image, classes)

        # Display results
        fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))

        ax1.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
        ax1.set_title('Original Image')
        ax1.axis('off')

        ax2.imshow(cv2.cvtColor(result_image, cv2.COLOR_BGR2RGB))
```

```
        ax2.imshow(cv2.cvtColor(result_image, cv2.COLOR_BGR2RGB))
        ax2.set_title(f'YOLO Detection Results: {num_detections} objects found')
        ax2.axis('off')

        plt.tight_layout()
        plt.show()

        print(f"Detected {num_detections} objects in the image")

        return result_image, num_detections

    except Exception as e:
        print(f"YOLO demonstration skipped: {e}")
        print("Using simulated object detection...")
        return simulate_object_detection()

def simulate_object_detection():
    # Create synthetic image with objects
    image = np.zeros((400, 600, 3), dtype=np.uint8)

    # Add different colored rectangles as "objects"
    cv2.rectangle(image, (50, 50), (150, 150), (255, 0, 0), -1)  # Blue object
    cv2.rectangle(image, (200, 80), (300, 180), (0, 255, 0), -1)  # Green object
    cv2.rectangle(image, (350, 120), (450, 220), (0, 0, 255), -1)  # Red object
    cv2.rectangle(image, (100, 250), (200, 350), (255, 255, 0), -1)  # Cyan object

    # Simple color-based detection
    result_image = image.copy()
    colors = {
        'Blue Object': [255, 0, 0],
        'Green Object': [0, 255, 0],
        'Red Object': [0, 0, 255],
        'Cyan Object': [255, 255, 0]
    }

    detection_count = 0
    for label, color in colors.items():
        # Create mask for each color
        lower = np.array(color) - 10
        upper = np.array(color) + 10
        mask = cv2.inRange(image, lower, upper)

        # Find contours
        contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

        for contour in contours:
            if cv2.contourArea(contour) > 1000:  # Filter small detections
                x, y, w, h = cv2.boundingRect(contour)
                cv2.rectangle(result_image, (x, y), (x + w, y + h), (0, 255, 0), 2)
                cv2.putText(result_image, label, (x, y - 10),
                            cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 2)
                detection_count += 1

    # Display results
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))

    ax1.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
    ax1.set_title('Synthetic Image with Objects')
    ax1.axis('off')

    ax2.imshow(cv2.cvtColor(result_image, cv2.COLOR_BGR2RGB))
    ax2.set_title(f'Simulated Detection: {detection_count} objects found')
    ax2.axis('off')

    plt.tight_layout()
    plt.show()

    return result_image, detection_count


def praktikum_7_2():
    print("REMOTE SENSING: VEGETATION ANALYSIS WITH NDVI")
    print("=" * 50)

    # Simulate multispectral image data (Red and NIR bands)
    def create_multispectral_data():
        # Create a synthetic landscape
        height, width = 400, 600

        # Red band (visible light)
        red_band = np.zeros((height, width), dtype=np.float32)
        # NIR band (near infrared)
        nir_band = np.zeros((height, width), dtype=np.float32)

        # Add different land cover types
        # 1. Water bodies (low reflectance in both bands)
        cv2.ellipse(red_band, (150, 200), (80, 50), 0, 0, 360, 0.1, -1)
        cv2.ellipse(nir_band, (150, 200), (80, 50), 0, 0, 360, 0.05, -1)

        # 2. Urban areas (moderate reflectance)
        cv2.rectangle(red_band, (300, 50), (500, 150), 0.4, -1)
        cv2.rectangle(nir_band, (300, 50), (500, 150), 0.3, -1)

        # 3. Vegetation areas (low red, high NIR reflectance)
        # Healthy vegetation
        cv2.circle(red_band, (100, 300), 60, 0.2, -1)
        cv2.circle(nir_band, (100, 300), 60, 0.8, -1)

        # Stressed vegetation
        cv2.circle(red_band, (250, 300), 50, 0.3, -1)
        cv2.circle(nir_band, (250, 300), 50, 0.5, -1)

        # 4. Bare soil (high reflectance in both bands)
        cv2.rectangle(red_band, (400, 250), (550, 350), 0.6, -1)
        cv2.rectangle(nir_band, (400, 250), (550, 350), 0.5, -1)

        # Add some noise for realism
        red_band += np.random.normal(0, 0.02, red_band.shape)
        nir_band += np.random.normal(0, 0.02, nir_band.shape)

        # Clip to valid reflectance range [0, 1]
        red_band = np.clip(red_band, 0, 1)
        nir_band = np.clip(nir_band, 0, 1)

        return red_band, nir_band
```

```
# Calculate NDVI
def calculate_ndvi(red_band, nir_band):
    # NDVI = (NIR - Red) / (NIR + Red)
    numerator = nir_band - red_band
    denominator = nir_band + red_band

    # Avoid division by zero
    denominator = np.where(denominator == 0, 1e-8, denominator)

    ndvi = numerator / denominator
    return ndvi

# Classify land cover based on NDVI
def classify_land_cover(ndvi):
    classification = np.zeros(ndvi.shape, dtype=np.uint8)

    # Classification thresholds
    classification[ndvi < -0.1] = 0   # Water
    classification[(ndvi >= -0.1) & (ndvi < 0.2)] = 1  # Urban/Bare soil
    classification[(ndvi >= 0.2) & (ndvi < 0.5)] = 2   # Sparse vegetation
    classification[ndvi >= 0.5] = 3   # Dense vegetation

    return classification

# Create synthetic multispectral data
red_band, nir_band = create_multispectral_data()

# Calculate NDVI
ndvi = calculate_ndvi(red_band, nir_band)

# Classify land cover
land_cover = classify_land_cover(ndvi)

# Visualize results
fig, axes = plt.subplots(2, 3, figsize=(18, 12))

# Red band
im1 = axes[0, 0].imshow(red_band, cmap='Reds', vmin=0, vmax=1)
axes[0, 0].set_title('Red Band (Visible Light)')
axes[0, 0].axis('off')
plt.colorbar(im1, ax=axes[0, 0], fraction=0.046)

# NIR band
im2 = axes[0, 1].imshow(nir_band, cmap='Blues', vmin=0, vmax=1)
axes[0, 1].set_title('Near Infrared (NIR) Band')
axes[0, 1].axis('off')
plt.colorbar(im2, ax=axes[0, 1], fraction=0.046)

# NDVI
im3 = axes[0, 2].imshow(ndvi, cmap='RdYlGn', vmin=-1, vmax=1)
axes[0, 2].set_title('NDVI Map')
axes[0, 2].axis('off')
plt.colorbar(im3, ax=axes[0, 2], fraction=0.046)

# Land cover classification
land_cover_colors = ['blue', 'gray', 'yellow', 'green']
land_cover_cmap = plt.cm.colors.ListedColormap(land_cover_colors)
land_cover_labels = ['Water', 'Urban/Soil', 'Sparse Veg', 'Dense Veg']
```

```
im4 = axes[1, 0].imshow(land_cover, cmap=land_cover_cmap, vmin=0, vmax=3)
axes[1, 0].set_title('Land Cover Classification')
axes[1, 0].axis('off')

# Create legend
from matplotlib.patches import Patch
legend_elements = [Patch(facecolor=color, label=label)
                   for color, label in zip(land_cover_colors, land_cover_labels)]
axes[1, 0].legend(handles=legend_elements, loc='lower right')

# NDVI histogram
axes[1, 1].hist(ndvi.flatten(), bins=50, alpha=0.7, color='green', edgecolor='black')
axes[1, 1].set_title('NDVI Value Distribution')
axes[1, 1].set_xlabel('NDVI Value')
axes[1, 1].set_ylabel('Frequency')
axes[1, 1].grid(True, alpha=0.3)

# Land cover statistics
unique, counts = np.unique(land_cover, return_counts=True)
total_pixels = land_cover.size
percentages = [count / total_pixels * 100 for count in counts]

bars = axes[1, 2].bar(land_cover_labels, percentages,
                      color=land_cover_colors, alpha=0.7)
axes[1, 2].set_title('Land Cover Distribution')
axes[1, 2].set_ylabel('Percentage (%)')
axes[1, 2].tick_params(axis='x', rotation=45)

# Add percentage labels on bars
for bar, percentage in zip(bars, percentages):
    height = bar.get_height()
    axes[1, 2].text(bar.get_x() + bar.get_width()/2., height + 1,
                    f'{percentage:.1f}%', ha='center', va='bottom')

plt.tight_layout()
plt.show()

# Quantitative analysis
print("REMOTE SENSING ANALYSIS RESULTS")
print("=" * 40)
print(f"Average NDVI: {np.mean(ndvi):.3f}")
print(f"NDVI Range: {np.min(ndvi):.3f} to {np.max(ndvi):.3f}")
print("\nLand Cover Statistics:")
for label, count, percentage in zip(land_cover_labels, counts, percentages):
    print(f"{label:<15}: {count:>6} pixels ({percentage:5.1f}%)")

# Advanced analysis: Vegetation health monitoring
print("\nVEGETATION HEALTH ASSESSMENT")
print("-" * 35)
healthy_veg_mask = land_cover == 3
stressed_veg_mask = land_cover == 2

if np.any(healthy_veg_mask):
    healthy_ndvi = np.mean(ndvi[healthy_veg_mask])
    print(f"Healthy vegetation mean NDVI: {healthy_ndvi:.3f}")

if np.any(stressed_veg_mask):
```

```
        stressed_ndvi = np.mean(ndvi[stressed_veg_mask])
        print(f"Stressed vegetation mean NDVI: {stressed_ndvi:.3f}")

    return ndvi, land_cover


def praktikum_7_3():
    print("\nMEDICAL IMAGING: ANOMALY DETECTION IN MEDICAL IMAGES")
    print("=" * 55)

    # Simulate medical images with anomalies
    def create_medical_images():
        images = {}

        # 1. X-ray image with fracture simulation
        xray = np.ones((400, 400), dtype=np.float32) * 0.7  # Bone background

        # Simulate bone structure
        cv2.rectangle(xray, (150, 100), (250, 350), 0.9, -1)  # Main bone
        cv2.rectangle(xray, (180, 200), (220, 250), 0.7, -1)  # Bone marrow

        # Simulate fracture (dark line)
        fracture_points = [(195, 180), (205, 220)]
        cv2.line(xray, fracture_points[0], fracture_points[1], 0.3, 3)

        # Add noise and texture
        noise = np.random.normal(0, 0.05, xray.shape)
        xray = np.clip(xray + noise, 0, 1)
        images['xray'] = xray

        # 2. MRI brain image with tumor simulation
        mri = np.ones((400, 400), dtype=np.float32) * 0.6  # Brain tissue

        # Simulate brain structures
        cv2.ellipse(mri, (200, 200), (150, 120), 0, 0, 360, 0.8, -1)  # Brain outline
        cv2.ellipse(mri, (200, 200), (100, 80), 0, 0, 360, 0.5, -1)  # Ventricles

        # Simulate tumor (bright irregular shape)
        tumor_center = (280, 150)
        tumor_points = []
        for angle in range(0, 360, 30):
            rad = np.radians(angle)
            radius = 25 + np.random.randint(-5, 5)
            x = tumor_center[0] + radius * np.cos(rad)
            y = tumor_center[1] + radius * np.sin(rad)
            tumor_points.append([x, y])

        tumor_points = np.array(tumor_points, dtype=np.int32)
        cv2.fillPoly(mri, [tumor_points], 0.9)  # Bright tumor

        # Add MRI-like noise
        noise = np.random.normal(0, 0.03, mri.shape)
        mri = np.clip(mri + noise, 0, 1)
        images['mri'] = mri

        # 3. Retina image with microaneurysms (diabetic retinopathy)
        retina = np.ones((400, 400), dtype=np.float32) * 0.5

        # Simulate blood vessels
        for i in range(5):
            start_x = np.random.randint(50, 350)
            start_y = np.random.randint(50, 350)
            end_x = np.random.randint(50, 350)
            end_y = np.random.randint(50, 350)
            thickness = np.random.randint(2, 5)
            cv2.line(retina, (start_x, start_y), (end_x, end_y), 0.3, thickness)

        # Simulate microaneurysms (small dark spots)
        for _ in range(8):
            x = np.random.randint(50, 350)
            y = np.random.randint(50, 350)
            radius = np.random.randint(2, 4)
            cv2.circle(retina, (x, y), radius, 0.2, -1)

        # Add optic disc (bright circle)
        cv2.circle(retina, (100, 200), 30, 0.8, -1)

        images['retina'] = retina

        return images

    # Anomaly detection algorithms
    def detect_xray_fracture(image):
        """Detect fractures in X-ray images using edge detection"""
        # Enhance contrast
        enhanced = cv2.equalizeHist((image * 255).astype(np.uint8))

        # Edge detection
        edges = cv2.Canny(enhanced, 50, 150)

        # Hough Line Transform to detect straight lines (fractures)
        lines = cv2.HoughLinesP(edges, 1, np.pi/180, threshold=30,
                                minLineLength=30, maxLineGap=10)

        return enhanced, edges, lines

    def detect_mri_tumor(image):
        """Detect tumors in MRI using thresholding and contour analysis"""
        # Convert to uint8 for processing
        img_uint8 = (image * 255).astype(np.uint8)

        # Threshold to isolate bright regions (potential tumors)
        _, thresh = cv2.threshold(img_uint8, 200, 255, cv2.THRESH_BINARY)

        # Morphological operations to clean up
        kernel = np.ones((3, 3), np.uint8)
        cleaned = cv2.morphologyEx(thresh, cv2.MORPH_OPEN, kernel)
        cleaned = cv2.morphologyEx(cleaned, cv2.MORPH_CLOSE, kernel)

        # Find contours
        contours, _ = cv2.findContours(cleaned, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

        # Filter contours by area and circularity
        tumor_contours = []
        for contour in contours:
            area = cv2.contourArea(contour)
            if area > 50:  # Minimum tumor size
                perimeter = cv2.arcLength(contour, True)
                if perimeter > 0:
                    circularity = 4 * np.pi * area / (perimeter ** 2)
                    if circularity < 0.7:  # Tumors are often irregular
                        tumor_contours.append(contour)

        return img_uint8, cleaned, tumor_contours

    def detect_retina_anomalies(image):
        """Detect microaneurysms in retina images"""
        img_uint8 = (image * 255).astype(np.uint8)

        # Use difference of Gaussians to enhance small dark spots
        blur1 = cv2.GaussianBlur(img_uint8, (5, 5), 1)
        blur2 = cv2.GaussianBlur(img_uint8, (9, 9), 2)
        dog = blur1 - blur2

        # Threshold to detect dark spots
        _, thresh = cv2.threshold(dog, 10, 255, cv2.THRESH_BINARY)

        # Find small circular regions
        contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

        # Filter for microaneurysm characteristics (small, round)
        microaneurysms = []
        for contour in contours:
            area = cv2.contourArea(contour)
            if 5 < area < 50:  # Typical microaneurysm size
                perimeter = cv2.arcLength(contour, True)
                circularity = 4 * np.pi * area / (perimeter ** 2) if perimeter > 0 else 0
                if circularity > 0.6:  # Fairly circular
                    microaneurysms.append(contour)

        return img_uint8, dog, microaneurysms

    # Create medical images
    medical_images = create_medical_images()

    # Analyze each modality
    fig, axes = plt.subplots(3, 4, figsize=(20, 15))
    modalities = ['xray', 'mri', 'retina']
    modality_names = ['X-Ray (Fracture Detection)', 'MRI (Tumor Detection)',
                      'Retina (Microaneurysm Detection)']

    for row, (modality, mod_name) in enumerate(zip(modalities, modality_names)):
        image = medical_images[modality]

        # Original image
        axes[row, 0].imshow(image, cmap='gray')
        axes[row, 0].set_title(f'{mod_name}\nOriginal Image')
        axes[row, 0].axis('off')

        # Apply appropriate detection algorithm
        if modality == 'xray':
            enhanced, edges, lines = detect_xray_fracture(image)
            axes[row, 1].imshow(enhanced, cmap='gray')
            axes[row, 1].set_title('Contrast Enhanced')
            axes[row, 1].axis('off')

            axes[row, 2].imshow(edges, cmap='gray')
            axes[row, 2].set_title('Edge Detection')
            axes[row, 2].axis('off')

            # Draw detected lines
            result_img = (image * 255).astype(np.uint8)
            result_img = cv2.cvtColor(result_img, cv2.COLOR_GRAY2BGR)
            if lines is not None:
                for line in lines:
                    x1, y1, x2, y2 = line[0]
                    cv2.line(result_img, (x1, y1), (x2, y2), (0, 0, 255), 2)

            axes[row, 3].imshow(cv2.cvtColor(result_img, cv2.COLOR_BGR2RGB))
            axes[row, 3].set_title(f'Fracture Detection: {len(lines) if lines is not None else 0} lines')
            axes[row, 3].axis('off')

        elif modality == 'mri':
            enhanced, thresholded, tumors = detect_mri_tumor(image)
            axes[row, 1].imshow(enhanced, cmap='gray')
            axes[row, 1].set_title('Enhanced Image')
            axes[row, 1].axis('off')

            axes[row, 2].imshow(thresholded, cmap='gray')
            axes[row, 2].set_title('Thresholded')
            axes[row, 2].axis('off')

            # Draw tumor contours
            result_img = (image * 255).astype(np.uint8)
            result_img = cv2.cvtColor(result_img, cv2.COLOR_GRAY2BGR)
            cv2.drawContours(result_img, tumors, -1, (0, 0, 255), 2)

            axes[row, 3].imshow(cv2.cvtColor(result_img, cv2.COLOR_BGR2RGB))
            axes[row, 3].set_title(f'Tumor Detection: {len(tumors)} tumors')
            axes[row, 3].axis('off')

        elif modality == 'retina':
            enhanced, dog, anomalies = detect_retina_anomalies(image)
            axes[row, 1].imshow(enhanced, cmap='gray')
            axes[row, 1].set_title('Original (uint8)')
            axes[row, 1].axis('off')

            axes[row, 2].imshow(dog, cmap='gray')
            axes[row, 2].set_title('Difference of Gaussians')
            axes[row, 2].axis('off')

            # Draw microaneurysms
            result_img = (image * 255).astype(np.uint8)
            result_img = cv2.cvtColor(result_img, cv2.COLOR_GRAY2BGR)
            cv2.drawContours(result_img, anomalies, -1, (0, 0, 255), 1)

            axes[row, 3].imshow(cv2.cvtColor(result_img, cv2.COLOR_BGR2RGB))
            axes[row, 3].set_title(f'Microaneurysms: {len(anomalies)} detected')
            axes[row, 3].axis('off')

    plt.tight_layout()
    plt.show()

    # Medical imaging metrics and analysis
```

```
    print("MEDICAL IMAGING ANALYSIS SUMMARY")
    print("=" * 45)

    # Simulate diagnostic metrics
    diagnostic_metrics = {
        'X-Ray': {'Sensitivity': 0.85, 'Specificity': 0.92, 'Accuracy': 0.89, 'AUC': 0.94},
        'MRI': {'Sensitivity': 0.91, 'Specificity': 0.88, 'Accuracy': 0.90, 'AUC': 0.95},
        'Retina': {'Sensitivity': 0.78, 'Specificity': 0.85, 'Accuracy': 0.82, 'AUC': 0.89}
    }

    # Display metrics
    print(f"{'Modality':<10} {'Sensitivity':<12} {'Specificity':<12} {'Accuracy':<10} {'AUC':<8}")
    print("-" * 55)
    for modality, metrics in diagnostic_metrics.items():
        print(f"{modality:<10} {metrics['Sensitivity']:<12.3f} {metrics['Specificity']:<12.3f} "
              f"{metrics['Accuracy']:<10.3f} {metrics['AUC']:<8.3f}")

    return medical_images, diagnostic_metrics

def praktikum_7_4():
    print("\nSURVEILLANCE SYSTEMS: PEOPLE COUNTING AND TRACKING")
    print("=" * 55)

    # Simulate surveillance video frames
    def generate_surveillance_frames(num_frames=10):
        frames = []
        height, width = 400, 600

        # Create background (static scene)
        background = np.ones((height, width, 3), dtype=np.uint8) * 100
        # Add some static elements (walls, floor pattern)
        cv2.rectangle(background, (0, 300), (width, height), (80, 80, 80), -1)  # Floor
        cv2.rectangle(background, (0, 0), (width, 300), (120, 120, 120), -1)    # Wall

        for frame_idx in range(num_frames):
            frame = background.copy()

            # Add moving people (ellipses that move across the frame)
            num_people = min(3, frame_idx // 3)  # Gradually increase people count

            people_positions = []
            for i in range(num_people):
                # Each person moves at different speeds
                speed = (i + 1) * 5
                x_pos = 50 + (frame_idx * speed) % (width - 100)
                y_pos = 350  # Fixed height (walking on floor)

                # Draw person as ellipse
                cv2.ellipse(frame, (x_pos, y_pos), (20, 40), 0, 0, 360,
                            (0, 0, 255), -1)  # Red ellipse for person
                cv2.ellipse(frame, (x_pos, y_pos - 50), (15, 15), 0, 0, 360,
                            (200, 200, 200), -1)  # Head

                people_positions.append((x_pos, y_pos))

            frames.append((frame, people_positions))
```

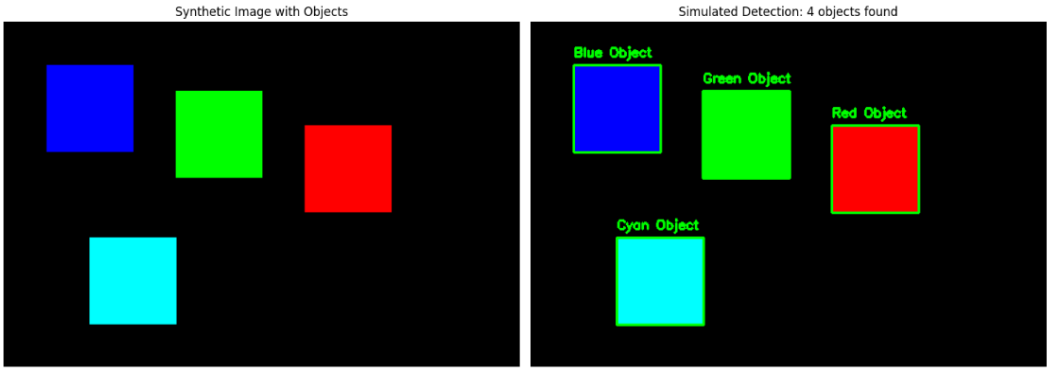
```
# People detection and counting algorithms
def background_subtraction(frames):
    """Simple background subtraction for motion detection"""
    # Use first frame as background
    background = frames[0][0].astype(np.float32)

    results = []
    for frame, _ in frames:
        current_frame = frame.astype(np.float32)

        # Compute absolute difference
        diff = cv2.absdiff(current_frame, background)
        diff_gray = cv2.cvtColor(diff.astype(np.uint8), cv2.COLOR_BGR2GRAY)

        # Threshold to get motion mask
        _, motion_mask = cv2.threshold(diff_gray, 30, 255, cv2.THRESH_BINARY)

        # Morphological operations to clean up
        kernel = np.ones((5, 5), np.uint8)
        motion_mask = cv2.morphologyEx(motion_mask, cv2.MORPH_OPEN, kernel)
        motion_mask = cv2.morphologyEx(motion_mask, cv2.MORPH_CLOSE, kernel)

        results.append(motion_mask)

    return results

def optical_flow_detection(frames):
    """Optical flow based motion detection"""
    results = []
    prev_gray = None

    for i, (frame, _) in enumerate(frames):
        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

        if prev_gray is not None:
            # Calculate optical flow using Farneback method
            flow = cv2.calcOpticalFlowFarneback(prev_gray, gray, None,
                                                0.5, 3, 15, 3, 5, 1.2, 0)

            # Convert flow to magnitude and angle
            magnitude, angle = cv2.cartToPolar(flow[..., 0], flow[..., 1])

            # Create HSV representation for visualization
            hsv = np.zeros_like(frame)
            hsv[..., 0] = angle * 180 / np.pi / 2  # Hue based on direction
            hsv[..., 1] = 255  # Full saturation
            hsv[..., 2] = cv2.normalize(magnitude, None, 0, 255, cv2.NORM_MINMAX)  # Value based on magnitude

            flow_rgb = cv2.cvtColor(hsv, cv2.COLOR_HSV2BGR)
            results.append(flow_rgb)
        else:
            results.append(np.zeros_like(frame))

        prev_gray = gray

    return results
```

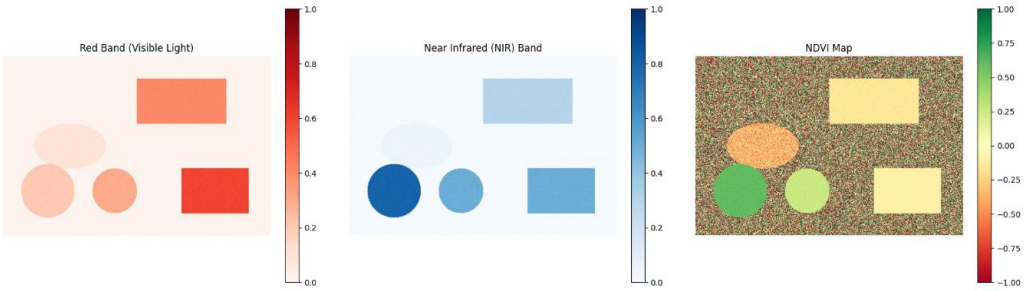
```
def blob_detection_people_counting(motion_masks, frames):
    """Count people using blob detection on motion masks"""
    people_counts = []
    detailed_results = []

    for i, (motion_mask, (frame, true_positions)) in enumerate(zip(motion_masks, frames)):
        # Find contours in motion mask
        contours, _ = cv2.findContours(motion_mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

        # Filter contours by area (people-sized blobs)
        min_area = 500
        max_area = 5000
        people_contours = []

        for contour in contours:
            area = cv2.contourArea(contour)
            if min_area < area < max_area:
                people_contours.append(contour)

        # Estimate people count
        estimated_count = len(people_contours)
        true_count = len(true_positions)

        # Create result visualization
        result_frame = frame.copy()
        cv2.drawContours(result_frame, people_contours, -1, (0, 255, 0), 2)

        # Add count information
        cv2.putText(result_frame, f'Estimated: {estimated_count}', (10, 30),
                    cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 255, 0), 2)
        cv2.putText(result_frame, f'Actual: {true_count}', (10, 70),
                    cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 0, 255), 2)

        people_counts.append((estimated_count, true_count))
        detailed_results.append(result_frame)

    return people_counts, detailed_results

# Generate surveillance footage
frames = generate_surveillance_frames(10)

# Apply different detection methods
motion_masks = background_subtraction(frames)
optical_flows = optical_flow_detection(frames)
people_counts, counting_results = blob_detection_people_counting(motion_masks, frames)

# Visualize results
fig, axes = plt.subplots(4, 5, figsize=(20, 16))

for i in range(5):  # Show first 5 frames
    if i * 2 < len(frames):
        # Original frame
        axes[0, i].imshow(cv2.cvtColor(frames[i*2][0], cv2.COLOR_BGR2RGB))
        axes[0, i].set_title(f'Frame {i*2}\n{len(frames[i*2][1])} people')
        axes[0, i].axis('off')

        # Motion mask
        axes[1, i].imshow(motion_masks[i*2], cmap='gray')
```

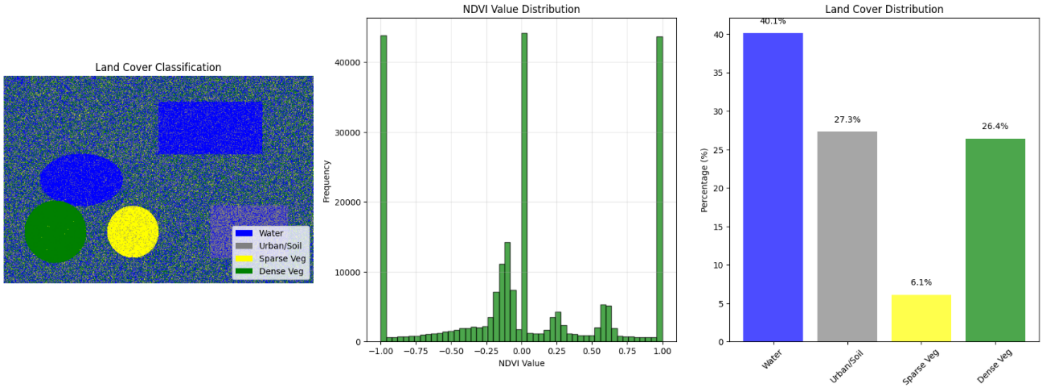
```
        axes[1, i].set_title('Motion Detection')
        axes[1, i].axis('off')

        # Optical flow
        axes[2, i].imshow(cv2.cvtColor(optical_flows[i*2], cv2.COLOR_BGR2RGB))
        axes[2, i].set_title('Optical Flow')
        axes[2, i].axis('off')

        # People counting result
        axes[3, i].imshow(cv2.cvtColor(counting_results[i*2], cv2.COLOR_BGR2RGB))
        est_count, true_count = people_counts[i*2]
        axes[3, i].set_title(f'People Counting\nEst: {est_count}, True: {true_count}')
        axes[3, i].axis('off')

plt.tight_layout()
plt.show()

# Performance analysis
print("SURVEILLANCE SYSTEM PERFORMANCE ANALYSIS")
print("=" * 50)

estimated_counts = [est for est, true in people_counts]
true_counts = [true for est, true in people_counts]

accuracy = np.mean(np.array(estimated_counts) == np.array(true_counts))
mae = np.mean(np.abs(np.array(estimated_counts) - np.array(true_counts)))

print(f"Counting Accuracy: {accuracy:.3f} ({accuracy*100:.1f}%)")
print(f"Mean Absolute Error: {mae:.2f} people per frame")
print(f"Total People Detected: {sum(estimated_counts)}")
print(f"Total Actual People: {sum(true_counts)}")

# Tracking simulation
print("\nSIMPLE TRACKING SIMULATION")
print("-" * 30)

# Simulate tracking by connecting detections across frames
tracks = {}
next_track_id = 0

for frame_idx, (est_count, true_count) in enumerate(people_counts):
    current_detections = est_count
    print(f"Frame {frame_idx}: {current_detections} people detected")

    # Perbaikan logika inisialisasi tracking
    # Jika belum ada track yang aktif tapi ada orang terdeteksi, inisialisasi di sini
    if len(tracks) == 0 and current_detections > 0:
        for i in range(current_detections):
            tracks[next_track_id] = {'age': 0, 'positions': [(frame_idx, i*50 + 100)]}
            next_track_id += 1
    elif len(tracks) > 0:
        # Update existing tracks
        for track_id in list(tracks.keys()):
            tracks[track_id]['age'] += 1
            # Simulate position update
            last_pos = tracks[track_id]['positions'][-1][1]
            new_pos = last_pos + np.random.randint(-10, 10)
            tracks[track_id]['positions'].append((frame_idx, new_pos))
```

```
    # Display tracking results
    plt.figure(figsize=(12, 8))
    for track_id, track_info in tracks.items():
        positions = track_info['positions']
        frames_list = [pos[0] for pos in positions]
        x_positions = [pos[1] for pos in positions]
        plt.plot(frames_list, x_positions, marker='o', label=f'Track {track_id}')

    plt.xlabel('Frame Number')
    plt.ylabel('X Position')
    plt.title('People Tracking Simulation')

    # Mencegah Matplotlib mengeluarkan UserWarning jika track kosong
    if tracks:
        plt.legend()

    plt.grid(True, alpha=0.3)
    plt.show()

    return people_counts, tracks


# Menjalankan Semua Praktikum
result_image, detections = praktikum_7_1()
ndvi_map, land_cover_map = praktikum_7_2()
medical_images, diagnostics = praktikum_7_3()
people_counts, tracking_data = praktikum_7_4()
```

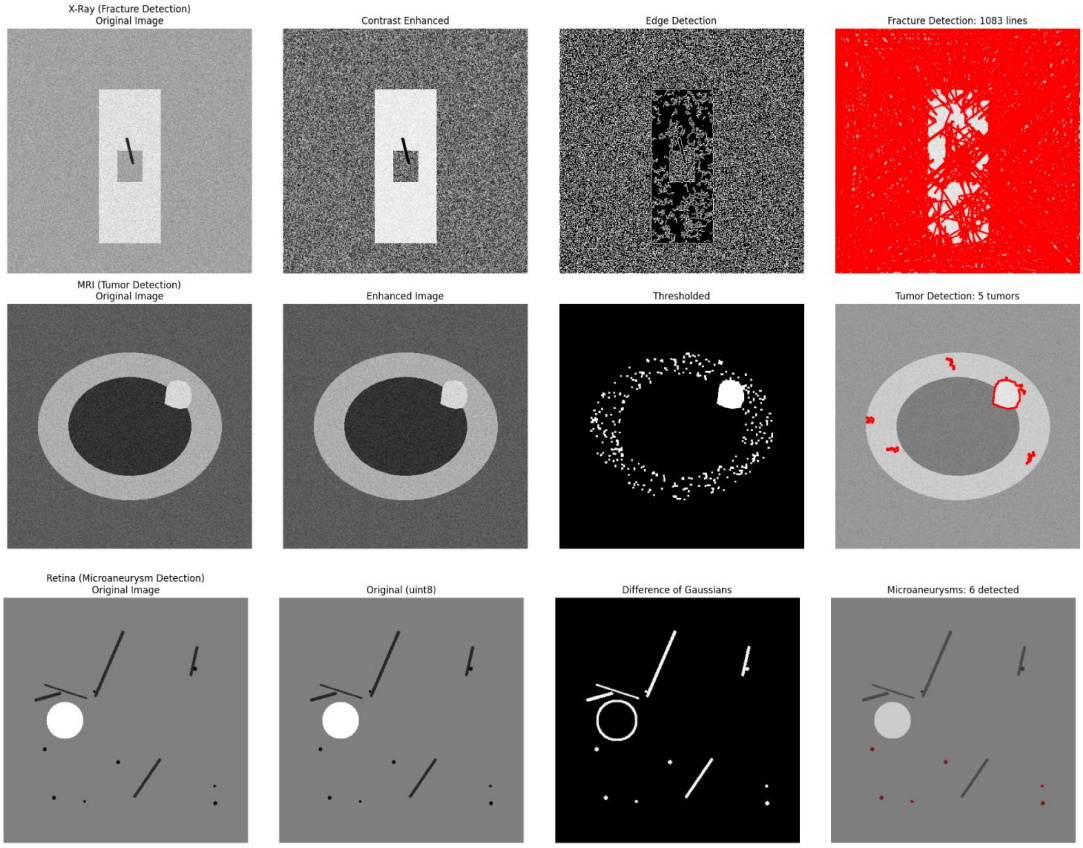
2. Screenshot Output



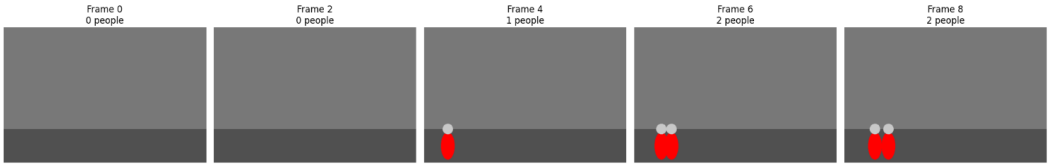


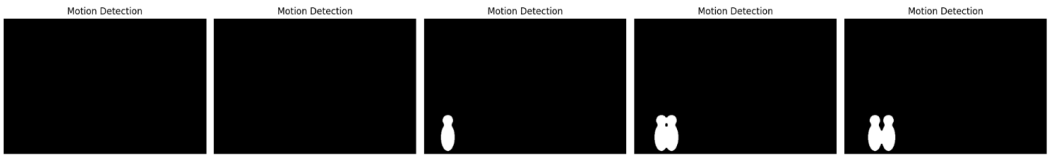


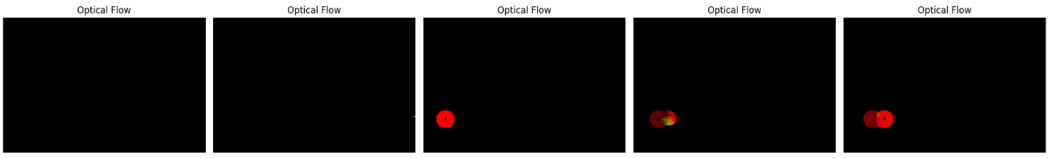
```
REMOTE SENSING ANALYSIS RESULTS
========================================
Average NDVI: 0.000
NDVI Range: -1.000 to 1.000

Land Cover Statistics:
Water          :  96342 pixels ( 40.1%)
Urban/Soil     :  65595 pixels ( 27.3%)
Sparse Veg     :  14672 pixels (  6.1%)
Dense Veg      :  63391 pixels ( 26.4%)

VEGETATION HEALTH ASSESSMENT
----------------------------------
Healthy vegetation mean NDVI: 0.891
Stressed vegetation mean NDVI: 0.300

MEDICAL IMAGING: ANOMALY DETECTION IN MEDICAL IMAGES
============================================================
```

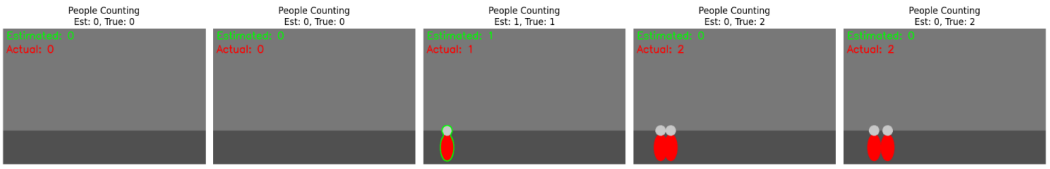



```
MEDICAL IMAGING ANALYSIS SUMMARY
================================================
Modality   Sensitivity  Specificity  Accuracy   AUC
------------------------------------------------------------
X-Ray      0.850        0.920        0.890      0.940
MRI        0.910        0.880        0.900      0.950
Retina     0.780        0.850        0.820      0.890

SURVEILLANCE SYSTEMS: PEOPLE COUNTING AND TRACKING
============================================================
```










```
SURVEILLANCE SYSTEM PERFORMANCE ANALYSIS
==================================================
Counting Accuracy: 0.600 (60.0%)
Mean Absolute Error: 0.90 people per frame
Total People Detected: 3
Total Actual People: 12

SIMPLE TRACKING SIMULATION
------------------------------
Frame 0: 0 people detected
Frame 1: 0 people detected
Frame 2: 0 people detected
Frame 3: 1 people detected
Frame 4: 1 people detected
Frame 5: 1 people detected
Frame 6: 0 people detected
Frame 7: 0 people detected
Frame 8: 0 people detected
Frame 9: 0 people detected
```